

Customer Churn Analysis – Summary

Project Objective

The objective of this project is to analyze customer behavior and identify the factors that contribute to customer churn. Understanding these factors can help businesses improve customer retention strategies and reduce revenue loss.

Dataset Overview

- Total Customers: **7043**
- Features: **21**
- Target Variable: **Churn**
 - Yes = Customer left the company
 - No = Customer stayed

Data preprocessing included:

- Handling missing values in the **TotalCharges** column.
 - Converting the **SeniorCitizen** column from 0/1 to No/Yes.
 - Checking for duplicates and null values.
-

Key Findings

1. Overall Churn Rate

- **73.46%** of customers stayed with the company.
- **26.54%** of customers churned.

This indicates that approximately **1 out of every 4 customers leaves the company**, which can significantly impact revenue and customer acquisition costs.

2. Churn by Gender

- Male and Female customers show almost similar churn behavior.
- Gender does not appear to be a strong factor influencing customer churn.

Conclusion: Customer retention strategies should focus on service and contract-related factors rather than gender.

3. Churn by Senior Citizen Status

Analysis shows that:

- Senior Citizens have a noticeably higher churn rate.

- Non-Senior Citizens are more likely to remain with the company.

Conclusion: Older customers may require additional support, personalized plans, or loyalty programs to reduce churn.

4. Churn by Customer Tenure

Customers with:

- **1–12 months tenure** show the highest churn rates.
- Long-term customers are significantly less likely to churn.

Conclusion: The first year is the most critical period for customer retention. Businesses should focus on onboarding and early customer engagement.

5. Churn by Contract Type

Customers with:

- **Month-to-Month Contracts** have the highest churn rate.
- **One-Year Contracts** have lower churn.
- **Two-Year Contracts** have the lowest churn rate.

Conclusion: Longer contract commitments improve customer retention significantly.

6. Churn by Services

Internet Service

- Customers using **Fiber Optic Internet** tend to churn more frequently.
- Customers using **DSL Internet** are more likely to stay.

Online Security

Customers without Online Security services show higher churn rates.

Online Backup

Customers without backup services churn more often.

Tech Support

Customers without technical support exhibit higher churn behavior.

Device Protection

Customers lacking device protection plans are more likely to leave.

Conclusion: Value-added services increase customer loyalty and reduce churn.

7. Streaming Services

- Customers using Streaming TV and Streaming Movies generally show better engagement.
- Customers without these services demonstrate slightly higher churn rates.

Conclusion: Additional entertainment services can improve customer retention.

8. Payment Method Analysis

Customers using:

- **Electronic Check** show the highest churn rates.
- Customers using automatic payment methods (Bank Transfer, Credit Card) are less likely to churn.

Conclusion: Customers enrolled in automated payment systems tend to remain longer with the company.

Business Recommendations

Recommendation 1: Focus on New Customers

Implement:

- Welcome offers
- Loyalty rewards
- Regular follow-ups during the first year

Recommendation 2: Promote Long-Term Contracts

Provide:

- Discounts for annual plans
- Exclusive benefits for long-term subscriptions

Recommendation 3: Bundle Value-Added Services

Encourage customers to subscribe to:

- Online Security
- Tech Support
- Online Backup
- Device Protection

Recommendation 4: Improve Senior Citizen Experience

Offer:

- Dedicated customer support
- Simplified plans
- Personalized assistance

Recommendation 5: Reduce Dependence on Electronic Checks

Encourage:

- Credit Card Auto-Pay
 - Bank Transfer Auto-Pay
-

Final Conclusion

The analysis reveals that customer churn is primarily influenced by:

1. Contract Type
2. Customer Tenure
3. Internet Service Type
4. Online Security & Tech Support Availability
5. Payment Method
6. Senior Citizen Status

Customers with month-to-month contracts, short tenure, electronic check payments, and without value-added services are significantly more likely to churn. By targeting these high-risk groups, the company can improve customer retention and reduce churn effectively.

Overall Churn Rate: 26.54%

Customer Retention Rate: 73.46%